

User Manual

Quickstart

A graphical user interface has been developed. Installation requires a copy of the executable file and BED files of chromosomal gene coordinates, both of which can be found on the GitHub and should be located in the same file. User should download exe file and allow approval if OS asks. Upon launching, it may take upwards of 30 seconds for the program to load depending on computer specifications.

The initial screen is shown in figure 1. Users should click on file -> open file to upload their BED file to the program. The ideogram will be automatically populated with the regions from the BED file, and the user can then save the ideogram using file -> save image or using the save icon found above the ideogram. Help will take the user to the GitHub repository for this project.

A loaded BED file is shown in figure 2 and 3. Figure 2 shows a 2-sample file, and figure 3 shows a 3-sample file with region overlaps.

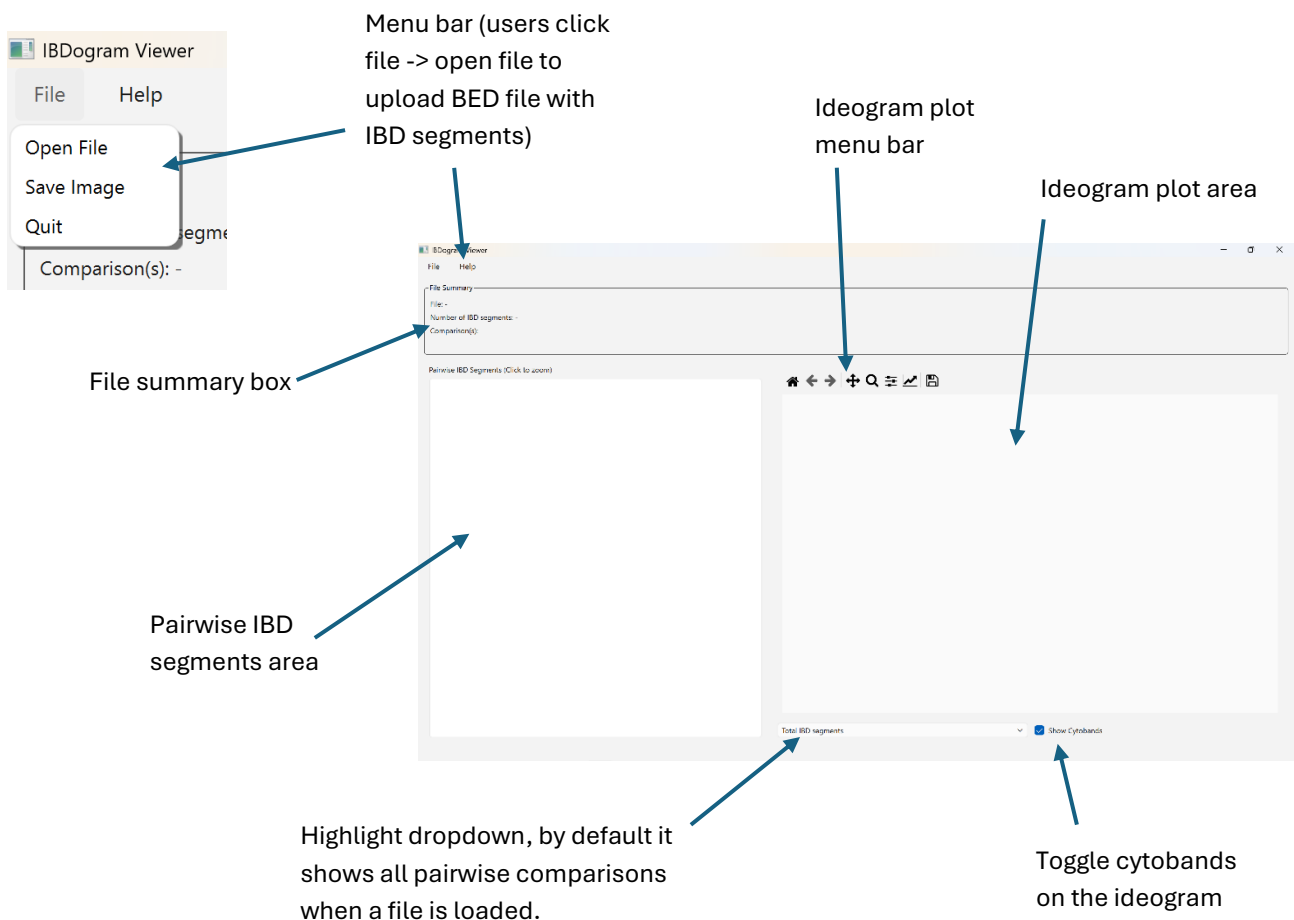
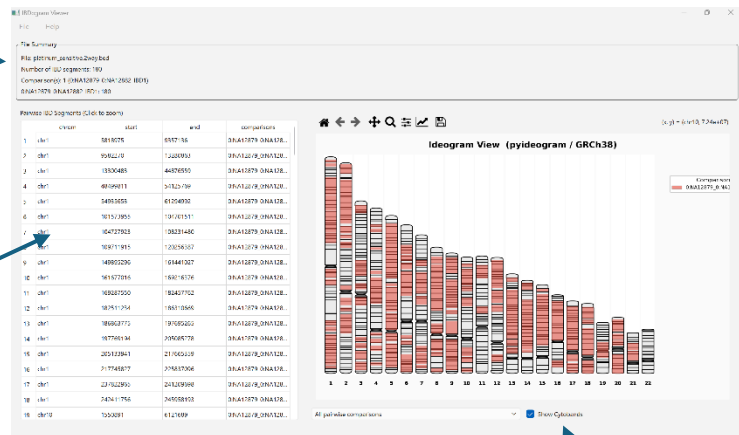


Figure 1. Screenshot of GUI without loaded file.

File summary including the filename, the number of IBD segments, and the number of comparisons

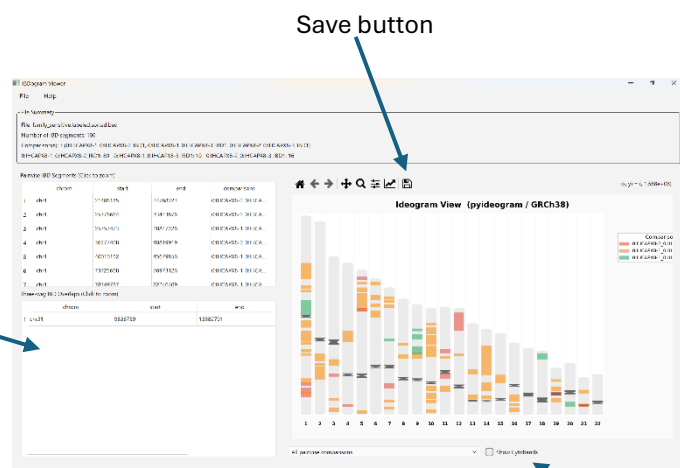
Pairwise IBD segments table (users can click on a row to call a popup of a specific region with a table of genes present in the region)



Toggle cytobands: ON

Figure 2. Two-way pairwise comparisons shown in GUI

Three way IBD overlaps table (users can click on a row to call a popup of the overlap region with a table of genes present in the region)



Toggle cytobands: OFF

Figure 3. Three-way pairwise comparisons shown in GUI

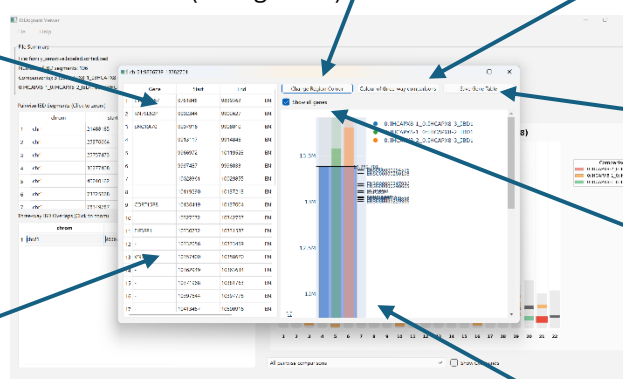
Gene table showing gene name, location, and ensemble ID

Change region colour (background)

Change colour of comparisons (legend)

Save gene table as CSV

Users can click on a row to select/deselect genes to appear on the region diagram when show all genes is toggled off



Toggle to show or hide all genes on the region diagram. Defaults to gene name otherwise uses Ensembl ID

Region diagram

Figure 4. Overlap region popup triggered by selecting a region in the main overlap table

Additional comments

The GUI has some functions that may require elaboration. In the main window, by default it will show all the pairwise comparisons on the ideogram. This can be changed using the dropdown menu to specify which comparison should be highlighted on the graph. There is a tick box to show or hide the cytobands to user preferences, and the centromeres (shown in grey) will not be toggled by the tick box. The ideogram image will be saved as a PNG regardless of which save button is used. If the user has multiple files to analyse, choosing file -> open file will automatically refresh the GUI to the new uploaded file without the user having to reopen the program.

Both tables present in the main window have the functionality to call a popup of the region when a row is clicked on by the user. When an IBD segment is clicked on from the pairwise comparison table, a popup window will appear showing genes within the IBD segment and illustrating their locations within the segment. There will be no overlaps shown in the diagram, just the region, and the button to change the colour of the comparisons will be unusable for the popup from the pairwise comparison table. A gene table will be shown in the popup for both the simple pairwise comparison regions and the overlapping regions and can be saved by the user as a CSV. The show all genes tick box defaults to on, but the user can toggle it off and select specific genes from the gene table to appear on the region diagram

For the overlapping regions, the diagram in the pop up will show the comparisons on top of the diagram so it is easier to visualize. It has the same functions as the pairwise comparison popup, though the comparison colour change button will allow the user to choose 3 colours and the diagram will change to reflect the user's choice.

Known points for improvement

In future releases of this program, it would be beneficial to investigate some changes. Loading the program can take some time, so future releases would investigate the cause and perhaps create a splash screen to feed back to users that the program is loading.

For the GUI itself, when the file is loaded it may be difficult to see the legend of the ideogram due to long sample names. The legend of the ideogram would be edited in future to determine size based on the length of the sample names, so images saved will include the whole sample name. This project could also be expanded to allow more relaxed overlaps, demonstrating the presence of an overlap between A-B and B-C but not A-C. In addition, the controls on top of the ideogram generated by matplotlib may be further optimised to specifics of ideogram content, or other controls could be added. An example of this would be the ability to specify which comparisons to show on the ideogram, if the user wanted to show two out of three, or to allow the user to change the colour of the highlight for each comparison in the file.

The region popup should be improved in the future with updates to the gene table, making it more intuitive so users can better see which genes have been selected on the graph, and a gene search bar if the user is interested in a specific gene. Improvements could also be made to user control over the popup window, including colour coding and window size management. In

addition, a save region functionality to save the image of the region could be helpful, especially when a user is only interested in specific genes.